

Cybersecurity

Linux Access Controls

Lesson 4.2



Learning Objectives

By the end of today's class, you will be able to:

- 1 Audit passwords using **john**.
- 2 Elevate privileges with **sudo** and **su**.
- 3 Create and manage users and groups.
- 4 Inspect and set file permissions for sensitive files on the system.



Welcome



Recap

Linux and Command Line

01

Linux history and distributions

02

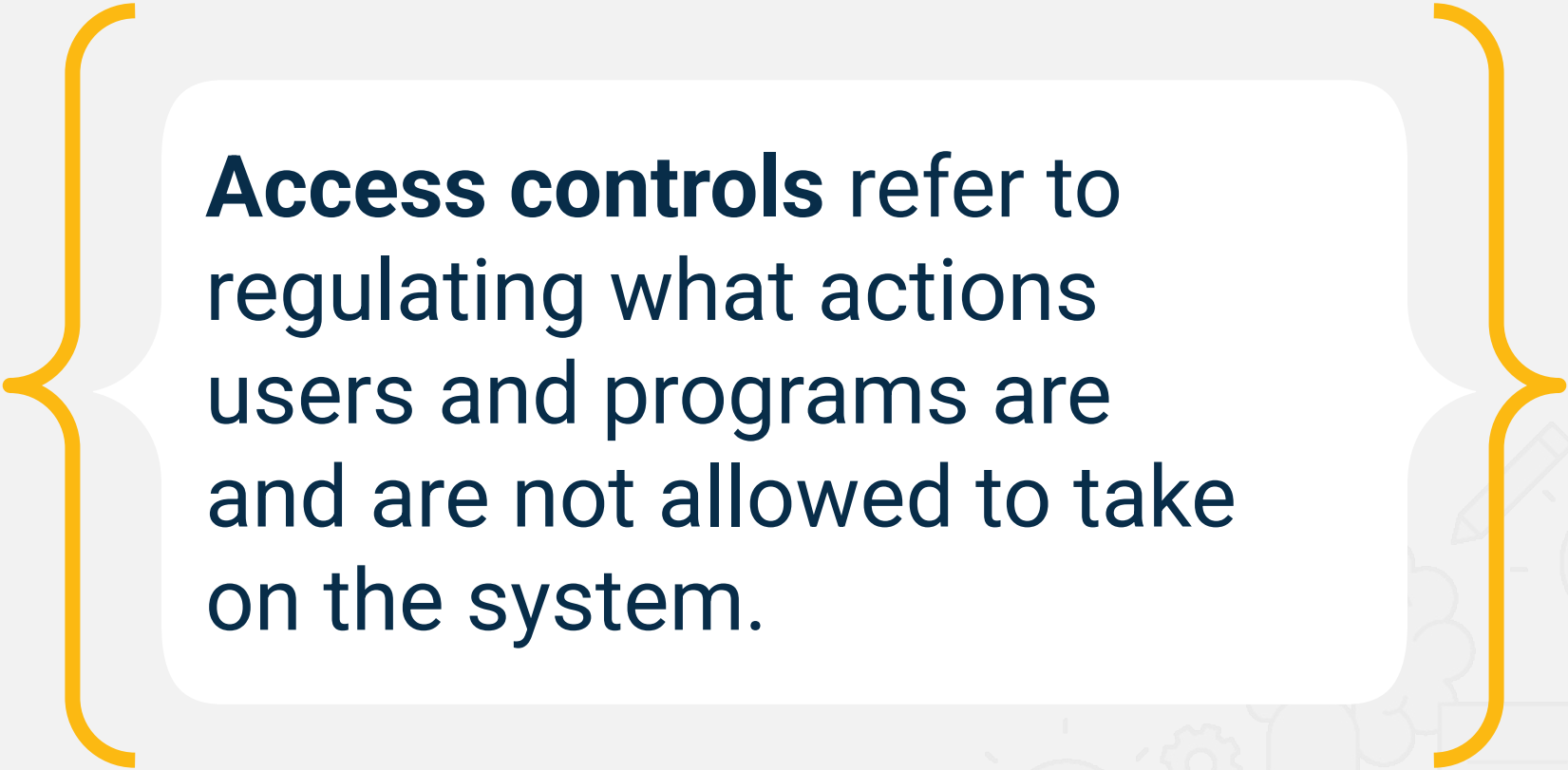
Navigating file structure with command line

03

Managing processes with **top**, **ps**, and **kill**

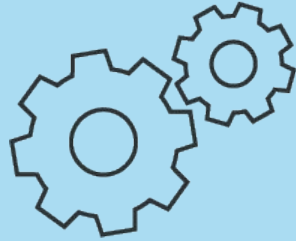
04

Installing packages with **apt**

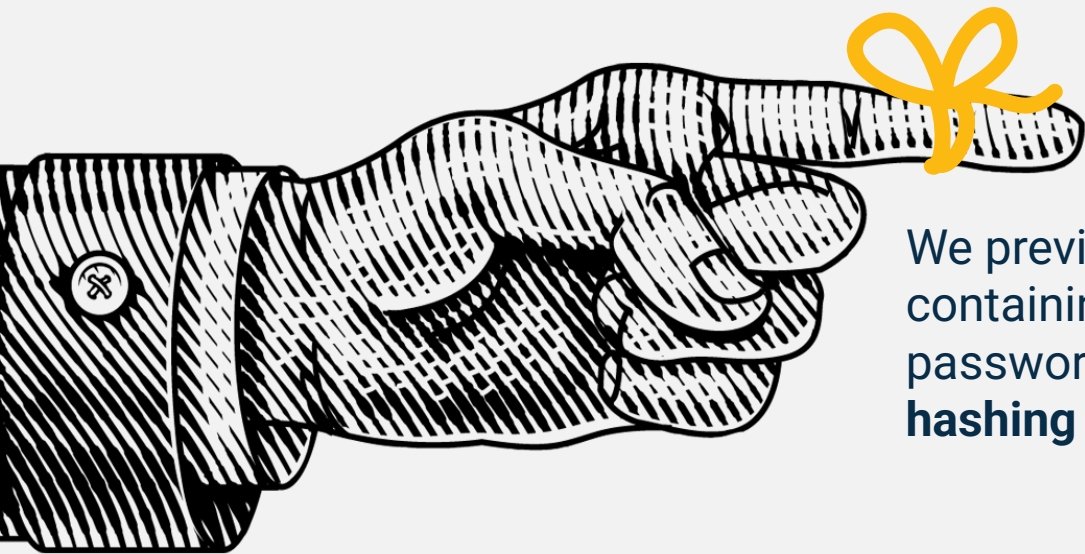


Access controls refer to regulating what actions users and programs are and are not allowed to take on the system.



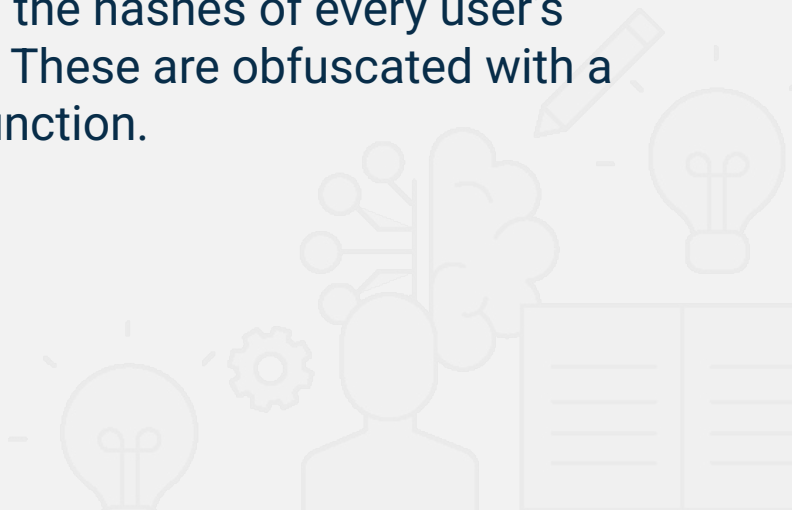


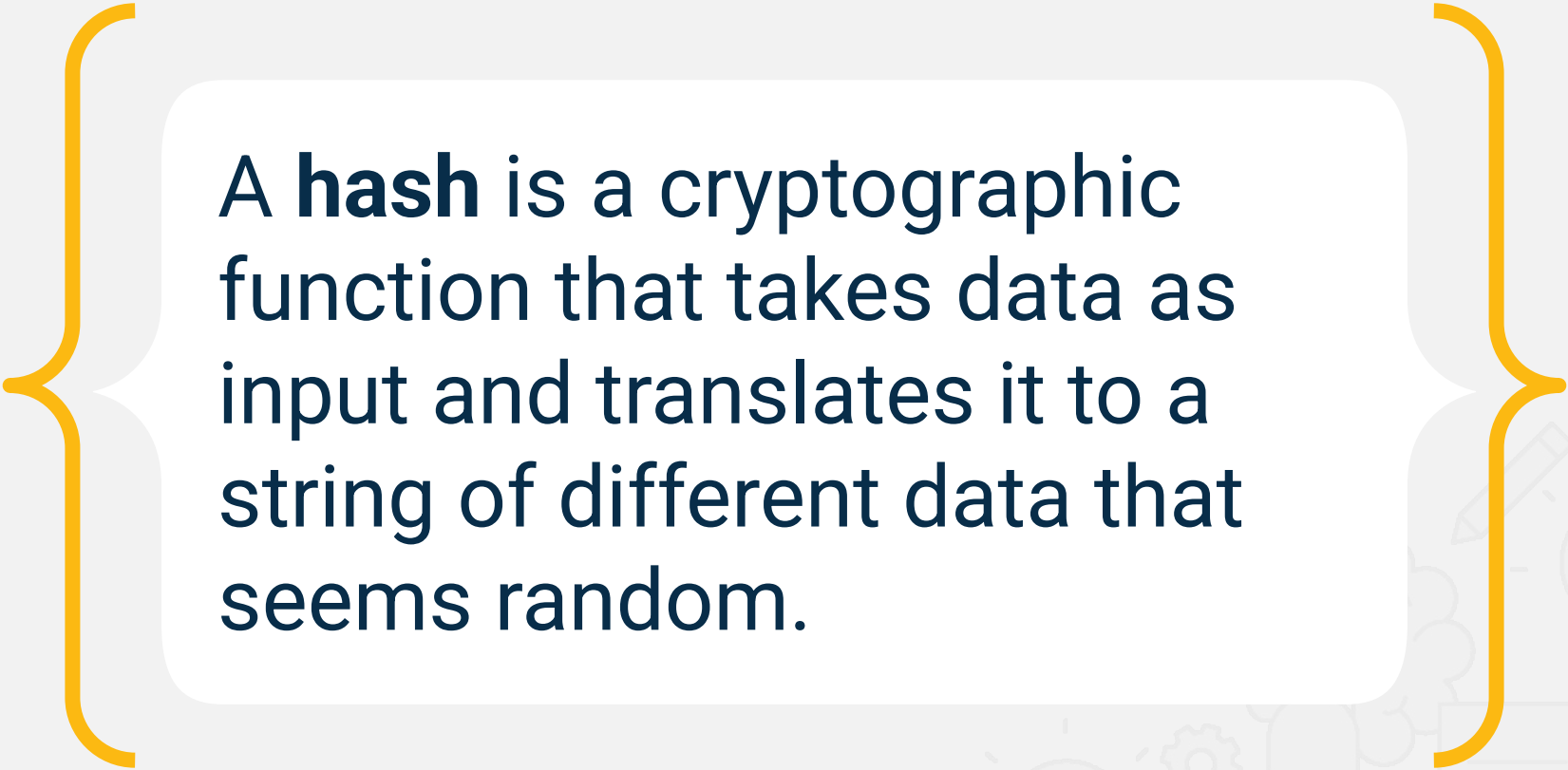
Users and Passwords



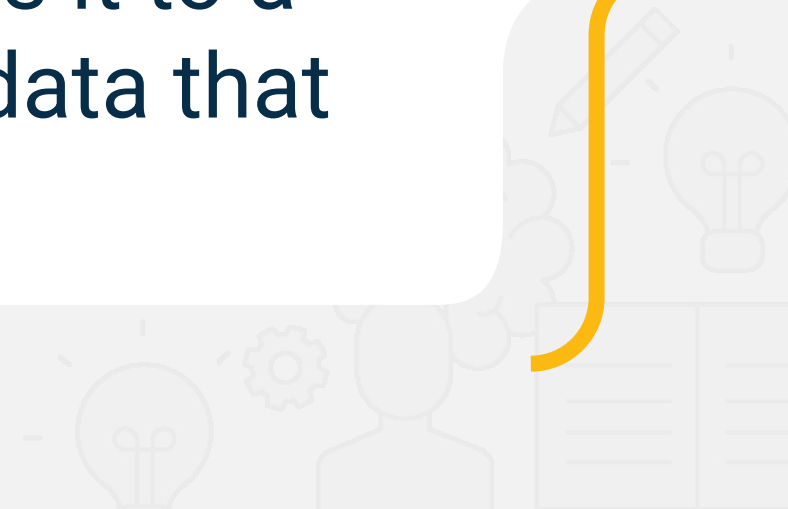
Remember,

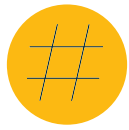
We previously examined the `/etc/shadow` file, containing the hashes of every user's password. These are obfuscated with a **hashing** function.





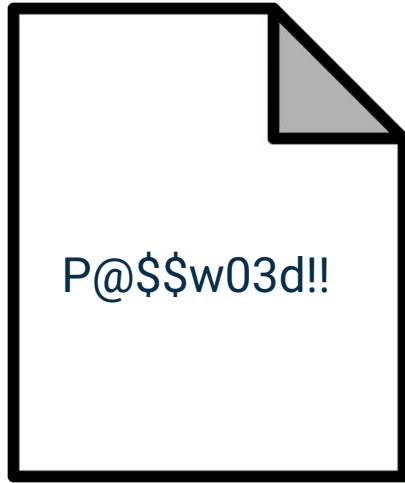
A **hash** is a cryptographic function that takes data as input and translates it to a string of different data that seems random.



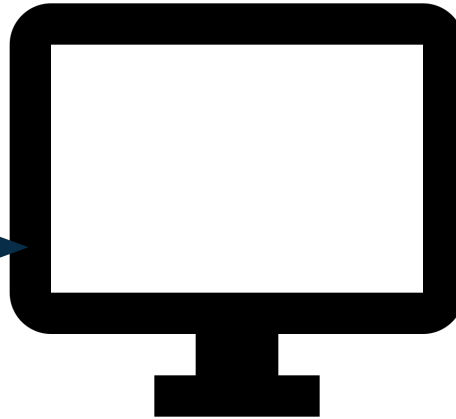


Password Hashes

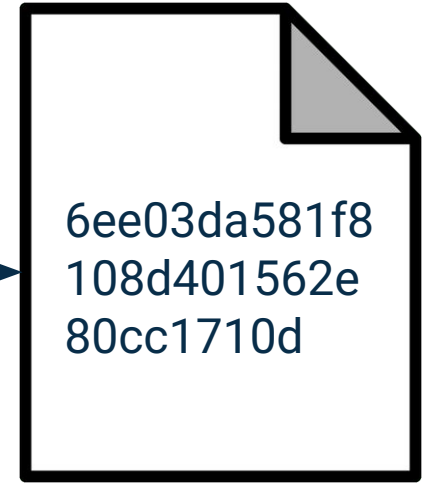
Plain Text Password



Hashing Function



Hashed Password

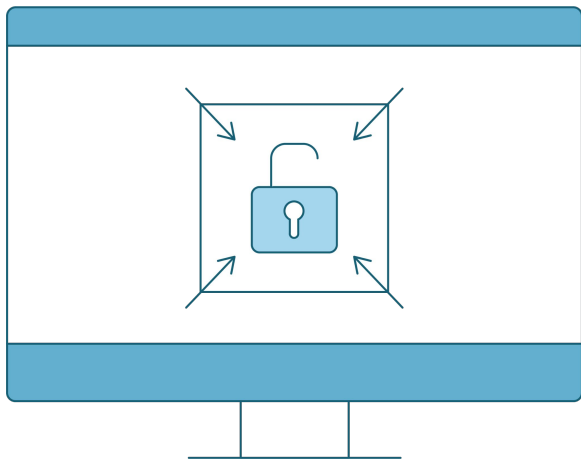


Remember: A hash will always output the same string for the same input data.

Password Cracking

Brute Force Attacks

Password Cracked



- Cracking tools do not reverse a password hash
 - Generate hashes of words/characters
 - Custom wordlists
 - Prebuilt popular wordlists
- Computed hashes compared to original hash
 - If two match, password has been found
- This is a form of **brute force attack**



What makes a stronger
password? Its
complexity or **length**?



What is the current
industry-standard length
for a password?



8 Characters



How Secure Is My Password?

Go to [How Secure Is My Password? | Password Strength Checker](#)

How Secure Is My Password?



The #1 Password Strength Tool. Trusted and used by millions.

A white rectangular input field with a light blue arrow icon on the left and a small grey icon on the right. The text "ENTER PASSWORD" is displayed in a light grey font inside the field.

Secure Password Takeaways

If a system requires passwords of only 16 characters, the password will be relatively strong, even if it contains words.

Add a few extra characters and it will become exponentially more secure.

In contrast, if a password uses all random characters, it still must be at least 10-characters long to be very effective.

Password

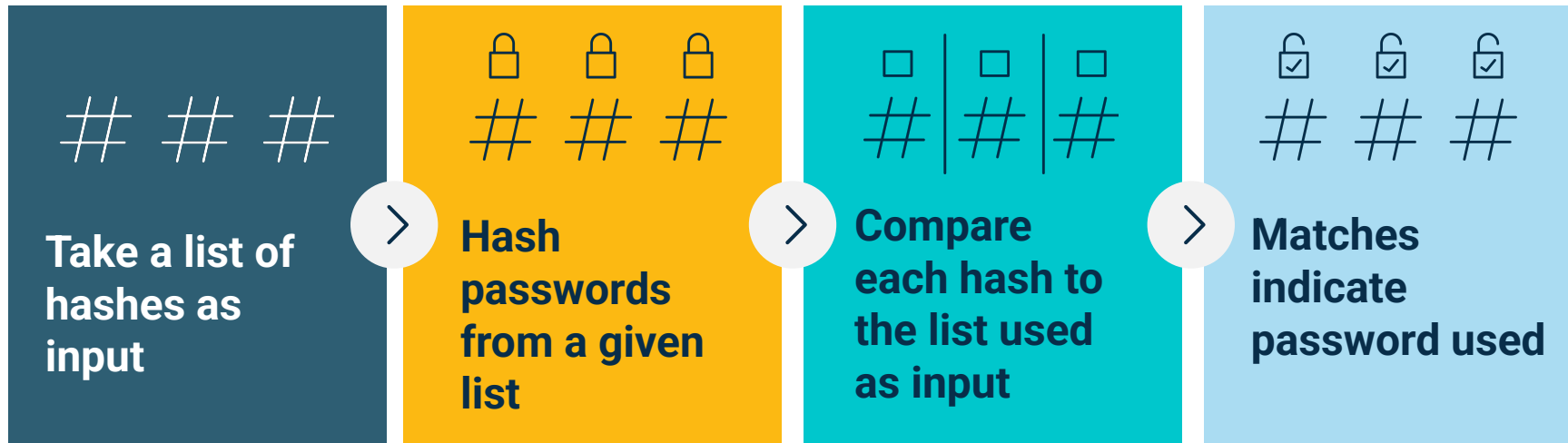
 Weak

Password

 Strong

Cracking Passwords

Modern password cracking software works using the following steps:



John the Ripper

Two-step process:

1. Create a hashlist
2. Run John the Ripper





Creating the Hashlist

John the Ripper

John can take an input file containing:

- Usernames
- Password hashes

John requires files to adhere to specific format:

```
admin:e08e4506d2e3f370a5e8ab79647df309
```

Username

Password Hash



Running John the Ripper

Syntax Options

Basic Usage

```
john <hashlist>
```

Pre-defined wordlist

```
john <hashlist> --wordlist=/path/to/wordlist
```

Cracked hashes

```
john --show <hashlist>
```



Activity:

Talk to John

In this activity, you will continue your role as a junior administrator auditing the system. Now, the focus will be on passwords.

- You will use john, which we installed in the previous class, to crack the password hashes for several of the users on our system.

Suggested Time:
20 Minutes

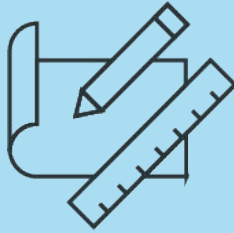


Time's up!
Let's review



Questions?





Privileges, root, sudo, and su



We've used `sudo` for several commands in this unit.

Why do you think some commands require `sudo` and others don't?

Privileges and Users

Why do we need to use sudo for some commands, and not others?

Users

Every file and program on a Linux system has permissions.

These permissions tell the system which users can access a file or run a program.

Groups

Users can be placed in groups, which can have their own permissions.

Root

File and program permissions apply to all users **except** the root.

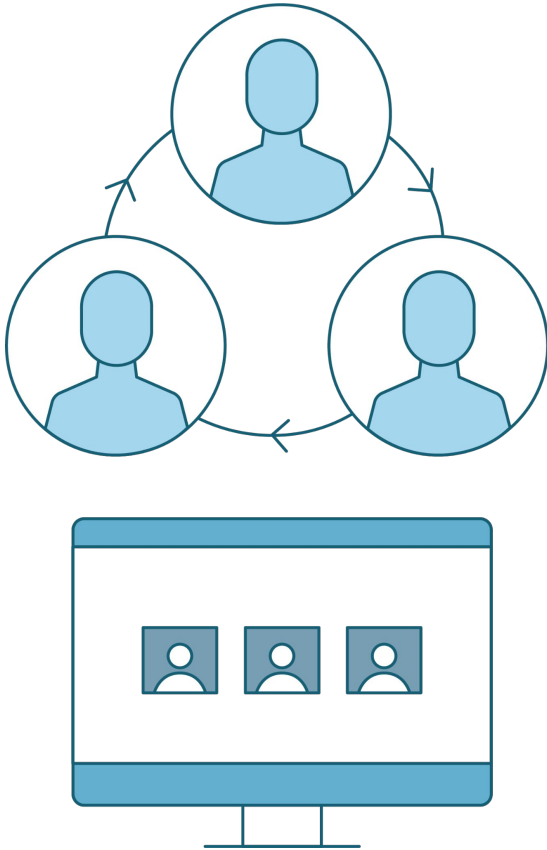
The root user (or super user) has complete access to the system and can perform any task.

Switching Users

Elevating Privileges

In Linux, you can access different users with **su**

- “Switch User”
- **su <username>**
- Helpful for troubleshooting
- Imitate the user



Root Access

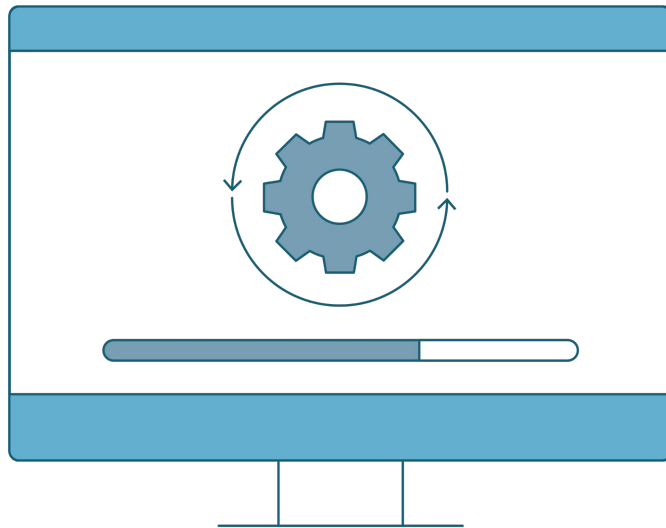
-  Properly secured Linux systems do not allow anyone to login as **root**
-  **sudo** stands for “**superuser do**”
-  After the **sudo** command is run, users are reverted to their usual access
-  **sudo** can also control which commands the user can run as root.

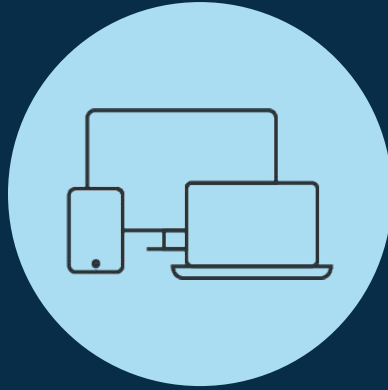


Demonstration Scenario

We will attempt to update all of our existing software packages.

- If our privileges do not allow us to do so, we will first use `su` to switch directly to the root user.
- We'll then demonstrate the dangers of working directly as the root user.
- We'll then do the same updates by using `sudo` instead and demonstrate why this is the more secure option.





Instructor Demonstration

su VS sudo



Demo Summary

Let's Recap



`whoami`



`sudo`



`su`



`sudo -l`



`visudo`



Activity:

sudo Wrestling

So far, we have been using administrator privileges to stop processes, install software, copy sensitive files, run programs, and edit configuration files.

- The senior administrator has asked that you audit the system for **sudo** and **root** access, making sure that no users other than the admin user have access to any **sudo** use.
- Lock access to the compromised system, including **sudo** access.

Suggested Time:
25 Minutes



Time's up!
Let's review



Questions?





Break
15 mins



Users and Groups

Users and Groups

Users on a Linux system can be added to groups. Linux has the ability to create groups of users for functions like file and service sharing.

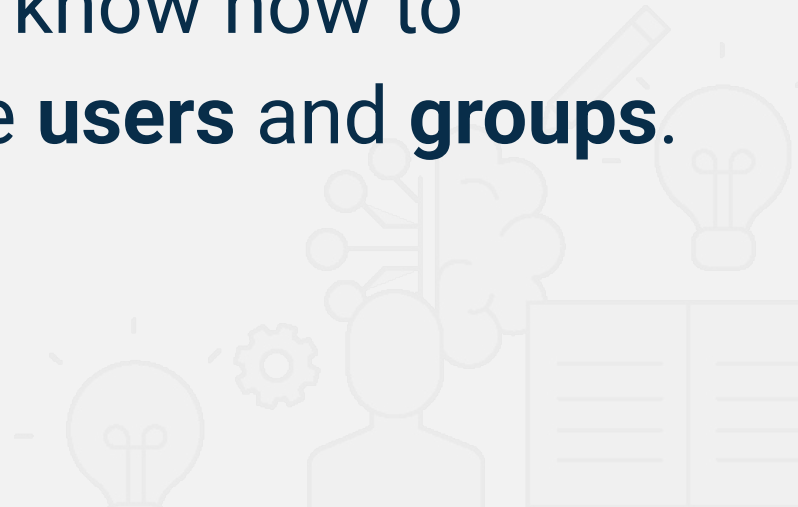
If a company has different departments, like sales, accounting, and marketing, a Linux admin can create a group for each department.

Only users in each group can access files owned by the group.





System administrators
need to know how to
manage **users** and **groups**.



id Command

Linux identifies users and groups in the system using the **id** command:

- 1 Linux associates a specific number with each user, known as the **user ID (UID)**.
- 2 Linux uses the **UID**, not the username.
- 3 System users have a UID that is **less than 1,000**.
- 4 Standard users have a UID that is **greater than 1,000**.
- 5 The root user always has the UID of **0**.



Demonstration Scenario

In the upcoming demo, we'll dive into more actions for user and group management using the following scenario:

Your company recently made changes to the developer team.



Mike, a lead developer, has left the company.



Joseph has joined as a new junior developer.



Demonstration Scenario

The company's Linux system has never been set up properly with a `developers` group. Instead, **Mike was part of the general group.**

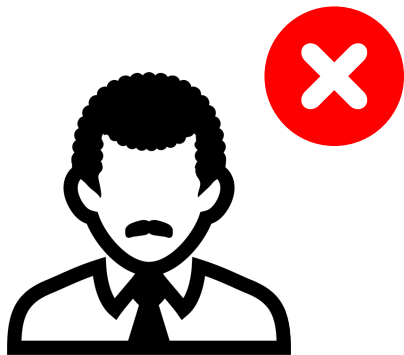
General group



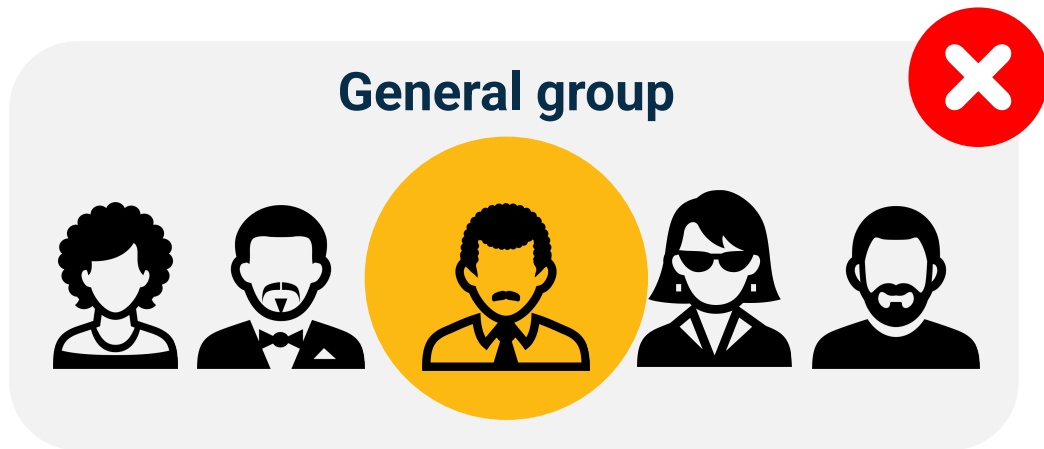


Demonstration Scenario

As the sysadmin for this system, you need to remove Mike from the general group, remove the general group, and delete Mike from the system.



Mike, a lead developer, has left the company.





Demonstration Scenario

Then, you need to add Joseph to the system, create a developers group, and add Joseph to this group.



Joseph has joined as a new junior developer.

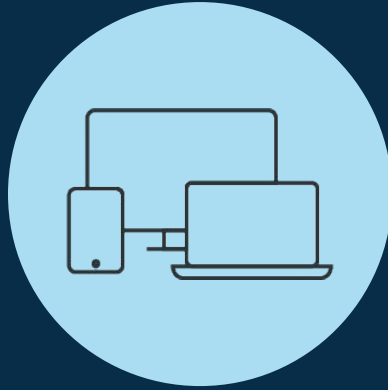
Developers group



Users and Groups Scenario

To accomplish these tasks, we will use the following commands:

groups	Get group info for the user <code>mike</code> .
usermod	Lock Mike's account to prevent him from logging in.
usermod	Remove the user <code>mike</code> from the <code>general</code> group.
<code>deluser --remove-home</code>	Delete the user <code>mike</code> . 
delgroup	Delete the <code>general</code> group.
adduser	Create the user <code>joseph</code> . 
addgroup	Create a <code>developer</code> group.
Usermod	Add the user <code>joseph</code> to the <code>developer</code> group.



Instructor Demonstration

Users and Groups



Activity:

Users and Groups

The administrator has asked that you audit all the users and groups on the system.

- You will create a new group for the standard users and remove users from the `sudo` group.
- In the previous activities, you found some malicious users. Now you will remove them from the system entirely.
- Being able to manage users and groups is an essential skill of a system administrator. Here, we will follow the principle of least privilege to make sure that the users on the system only have the access we want them to have and are only members of the correct groups.

Suggested Time:
20 Minutes



Time's up!
Let's review



Questions?





Recap

Linux Access Controls

01

Cracked password hashes with **john**

02

Escalated privileges with **su** and **sudo**

03

Managed users and groups



Next class

Inspect and set file permissions, monitor services, and assign users for services



Thank you